Near-Infrared Spectroscopy for Monitoring the *p*-Diethylbenzene Separation Process

HOEIL CHUNG,* MIN-SIK KU, JAEBUM LEE, and JAEBUM CHOO

NIR Project Team, SK Corporation, 110 Nam-Gu, Kosa-Dong, Ulsan, Korea (H.C., M.-S.K.); and Department of Chemistry, Hanyang University, Ansan 425-791, Korea (J.L., J.C.)

Near-infrared (NIR) spectroscopy has been successfully used for the monitoring of important components in the *p*-diethylbenzene (PDEB) separation process. The process is composed of mostly diethylbenzene isomers (*ortho, meta,* and *para*) and extractant (*p*-xylene), as well as other C_9–C_{11} aromatic hydrocarbons. Therefore, the major concern in using NIR spectroscopy in this process was the spectral resolution of NIR spectra among diethylbenzene isomers, since the molecular structures of each isomer were very similar. NIR spectral features of *o*-diethylbenzene (ODEB), *m*-diethylbenzene (MDEB), and PDEB showed considerable spectral differences in the 2100–2220 nm range. These combination bands originated from the combination of the =C–H stretch at 3100–3000 cm^{-1} and C=C ring stretch at 1600–1450 cm^{-1}. Characteristic C=C ring stretches of each isomers in the IR region result in selective and identifiable spectral features in the NIR region. Partial least-squares (PLS) regression was used to build each calibration model for ODEB, MDEB, PDEB, and *p*-xylene (PX). PLS calibration results of the four components showed excellent correlation with gas chromatography data. The combination region (2100–2500 nm) provided the important isomeric spectral information for PLS calibration since the absorption bands in this region were the most sensitive and selective.

Index Headings: **Near-infrared spectroscopy; NIR; Diethylbenzene Isomers; *p*-Diethylbenzene; PDEB; Partial least-squares; PLS**

INTRODUCTION

p-Diethylbenzene (PDEB) is one of the important petrochemicals serving as a desorbent in the *p*-xylene (PX) separation process and is a starting material for the manufacture of divinyl benzene. Typically PDEB is (1) isolated from a process stream composed of mostly diethylbenzene isomers (approximately 94%) and other C_9–C_{11} aromatic hydrocarbons by selective adsorption of PDEB on a solid absorbent, and (2) followed by extraction with a solvent.[1–4] To monitor and control the PDEB separation process optimally, one must accurately and rapidly measure the concentrations of not only each diethylbenzene isomer but also extractant PX. The conventional analytical method is gas chromatography (GC), which requires a long analysis time (approximately 40 min) due to separation of very similar hydrocarbons. Additionally, one must carefully maintain GC conditions to achieve reliable analytical data since the diethylbenzene isomers are not always baseline separated from other peaks in the chromatogram.

Vibrational spectroscopic methods, especially near-infrared (NIR) spectroscopy,[5,6] are excellent quantitative analytical methods for the measurement of concentrations of important components in the PDEB separation process because they allow a rapid and simultaneous measurement with high reproducibility. The major concern of using NIR spectroscopy for the quantification of diethylbenzene isomers is the resolution of NIR spectra of each isomer. This is because the molecular structures of each isomer are very similar, while absorption bands in NIR are typically broad and less characterized compared to fundamental vibrations in the MIR (mid-infrared). Examination of NIR spectral features of *o*-diethylbenzene (ODEB), *m*-diethylbenzene (MDEB), and PDEB reveals that considerable spectral differences are observed at 2100–2220 nm, the combination of the =C–H stretch at 3100–3000 cm^{-1} and C=C ring stretch at 1600–1450 cm^{-1}. In order to understand the origin of the differences in this region, the frequency calculations were performed by using the quantum mechanical density functional theory (DFT) method[7–9] to clearly identify the characteristic fundamental bands of three isomers from a theoretical point of view. This approach was very helpful to explain the spectral differences of three isomers in the 2100–2220 nm region. Characteristic C=C ring stretches of each isomer in the MIR region resulted in selective and identifiable spectral features in the NIR region.

Partial least-squares (PLS) regression[10,11] was utilized to differentiate the spectral difference of ODEB, MDEB, PDEB, and PX, and quantify each component. For the development of PLS calibration models for these major components in the PDEB separation process, actual samples were collected from the pilot-scale simulated moving bed (SMB), which selectively separated PDEB. The mixtures were primarily composed of four components: ODEB, MDEB, PDEB, and PX. Minor C_9–C_{11} hydrocarbons are present in these mixtures. Overall, the samples were composed of 25–35 different hydrocarbons, which were primarily diethylbenzene isomers and PX. PLS calibration results of the four components showed excellent correlation with data from GC, which has been replaced by NIR spectroscopy for the measurement of important components in the PDEB separation process.

EXPERIMENTAL

Sample Preparation. One hundred fifty-two samples were obtained from the actual separation pilot plant (so-called simulated moving bed). This pilot plant was designed to operate under various conditions; therefore, concentrations of collected samples were widely varied. The concentration ranges of *o*-diethylbenzene, *m*-diethylbenzene, *p*-diethylbenzene, and *p*-xylene are summarized in Table I. As shown, the concentrations of each component vary widely. Immediately after collection, samples were stored in a refrigerator at 4 °C to prevent evap-

Received 29 November 1999; accepted 19 January 2000.
* Author to whom correspondence should be sent.

0003-7028/00/5405-0715$2.00/0
© 2000 Society for Applied Spectroscopy

TABLE I. The concentration variations of each component in the data set (volume %).

Component	Maximum	Minimum	Mean	Standard deviation
o-Diethylbenzene	4.24	0.00	1.13	1.15
m-Diethylbenzene	56.27	0.04	14.58	14.66
p-Diethylbenzene	76.15	0.15	19.28	18.68
p-Xylene	99.38	0.47	62.86	26.38

oration of the hydrocarbons. The concentrations of each component were measured by a Hewlett-Packard 5890 GC equipped with a cross-linked polyethylene glycol–terephthalic acid (PEG-TPA) column and flame ionization detector (FID). Pure PX and PDEB were obtained from Ulsan refinery at SK Corporation (Ulsan, Korea). MDEB (95% purity) was obtained by concentrating 60% MDEB solution by fractional distillation. ODEB was purchased from Fluka.

Collection of Spectra. NIR spectra were collected over the 1100 to 2500 nm spectral region with a NIRSystems Model 6500 spectrometer (Foss NIRSystems Inc., Silver Spring, MD) equipped with a quartz halogen lamp and PbS detector. The resolution of collected spectra was 10 nm with 2 nm data intervals. NIR spectra were obtained by transferring each sample into a quartz cuvette having a 0.5 mm pathlength. Samples were allowed to equilibrate to room temperature (22 °C), and then corresponding single-beam spectra were collected. Background spectra of air were collected for every sample immediately before the scan. A total of 152 absorbance spectra were collected. Each NIR sample spectrum consisted of an average of 16 scans (acquisition time: 10 s).

MIR spectra (3500 to 600 cm^{-1}) of pure components were collected by using a Bio-Rad FTS-6000 FT-IR spectrometer equipped with a KBr beamsplitter and MCT detector. Each IR spectrum corresponded to an accumulation of 128 scans with a resolution of 2 cm^{-1}.

Data Processing. The 152 NIR spectra were divided into 106 spectra (70% randomly chosen) for calibration and 46 for confirmation. All data processing, including second-derivative and PLS regression, was accomplished by using NSAS Ver. 3.3 software (Foss NIRSystems Inc.).

RESULTS AND DISCUSSION

Near-Infrared Spectral Features. Selective quantification of each component, especially those with structural isomers, requires that at least minute qualitative spectral differences between each component be identified in the NIR region. Since NIR spectroscopy is not a sensitive technique in terms of structural change compared to other vibrational techniques such as mid-infrared (MIR) and Raman spectroscopy; it is necessary to initially examine the spectral features of major components in the PDEB separation process. NIR spectra (1100–2500 nm) of ODEB, MDEB, PDEB, and PX are shown Fig. 1. Also the 2100–2500 nm range is enlarged in the same plot. As expected, both the molar absorptivity and spectral selectivity decrease as the absorption bands are closer to the shorter wavelength region (further from the fundamental vibration in the MIR region). Generally for hydrocarbons, the 2100–2500, 1650–1800, 1300–1450, and 1100–1300

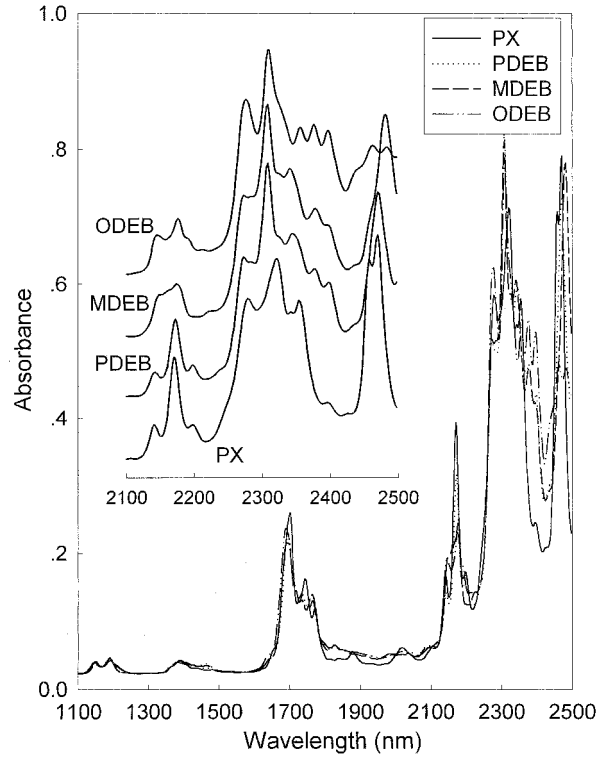

FIG. 1. NIR spectra (1100–2500 nm) of *o*-diethylbenzene (ODEB), *m*-diethylbenzene (MDEB), *p*-diethylbenzene (PDEB), and *p*-xylene (PX). The 2100–2500 nm range is expanded in the same plot.

bands correspond to the combination, first-overtone, combination, and second-overtone bands, respectively. The 2100–2500 nm region provides the best sensitivity in the NIR region.

Examination of the ODEB, MDEB, and PDEB spectra in Fig. 1 reveals that overall spectral features are slightly different, while the most qualitative differences are observed in the 2100–2220 nm region. In order to understand the origin of the spectral differences in this region, the MIR spectra were also collected and analyzed. The NIR and MIR spectra, which show the characteristic band patterns for each diethylbenzene isomers, are shown in Fig. 2. The 2100–2220 nm NIR bands result from the combination of =C–H stretch at 3100–3000 cm^{-1} and C=C ring stretch at 1600–1450 cm^{-1} (e.g., 3050 cm^{-1} + 1550 cm^{-1} = 4600 cm^{-1} = 2173 nm). In the =C–H stretching region, all three isomers show similar band

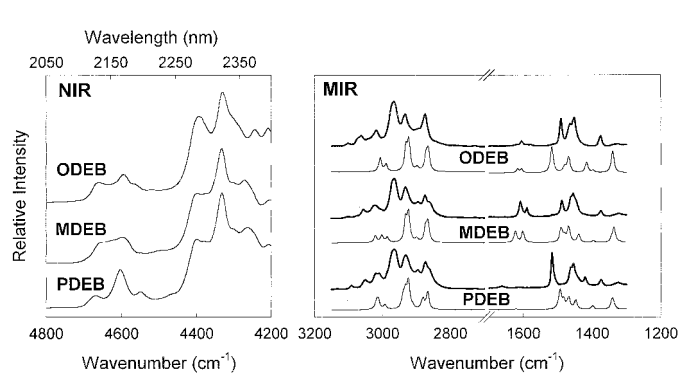

FIG. 2. NIR (2100–2500 nm), MIR (thick line), and simulated MIR (thin line) spectra of each diethylbenzene isomer.

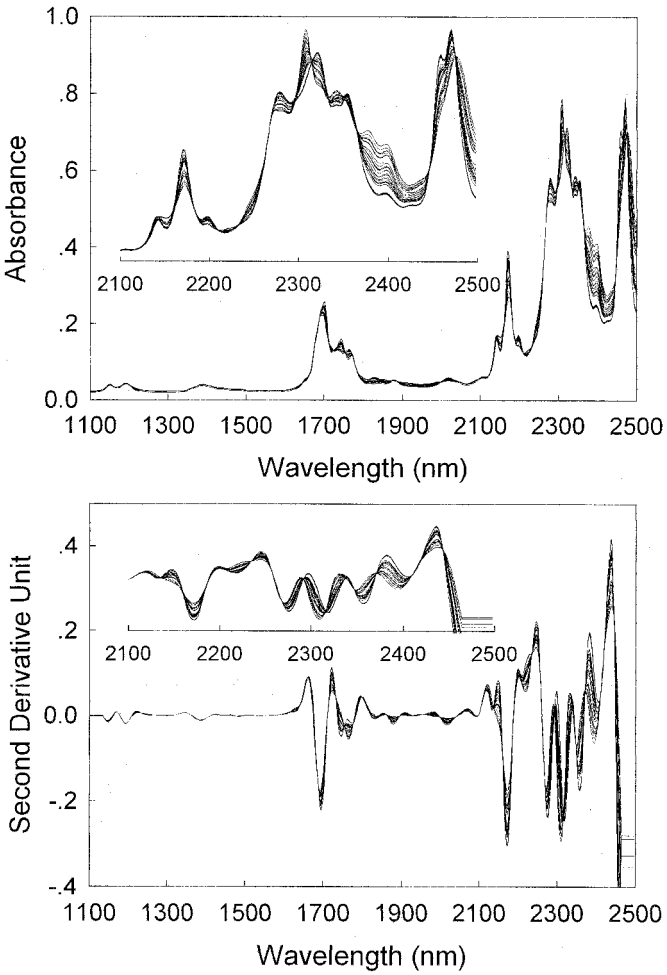

FIG. 3. NIR spectra of 40 randomly selected samples in the calibration set. Top and bottom plots correspond to log $1/R$ and second-derivative spectra, respectively.

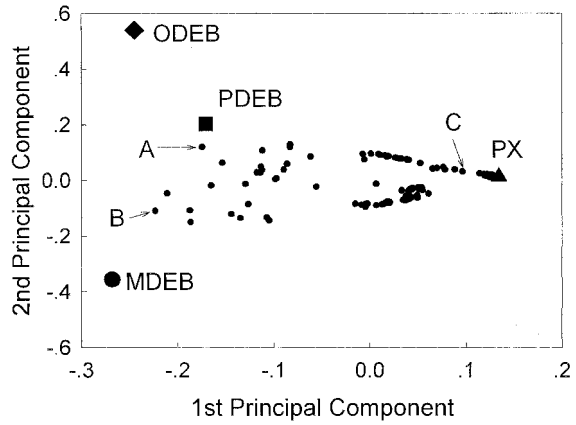

FIG. 4. Score scatter plot of calibration data set with pure components using first and second principal components. The scores of each pure component are presented as diamond (ODEB), circle (MDEB), square (PDEB), and triangle (PX).

shapes and intensities. However, aromatic C=C stretching band patterns between 1610 and 1450 cm^{-1} are very different; therefore, the relative band shapes and number of bands in this region can be used to distinguish the three isomers. For all three isomers, similar patterns of C=C stretching doublets are observed between 1465 and 1450 cm^{-1}. The next C=C stretching band is observed at 1490 cm^{-1} with moderate intensity for ODEB and MDEB, while a strong band is observed at 1516 cm^{-1} for PDEB. For ODEB and MDEB, additional C=C stretching bands are observed in the 1610–1570 cm^{-1} region; however, no additional band is observed in the 1610–1570 cm^{-1} region for PDEB.

In order to deduce the different band properties in the C=C stretching region, simulated theoretical infrared spectra were generated with the use of B3LYP/6-31G* calculations implemented in the *Gaussian94* program.[12] The simulated spectra of three isomers are also shown in Fig. 2 (thin line), and the differences in the C=C stretching bands are clearly recognized. According to the calculations, optimized ODEB and MDEB has C_2 symmetries, while PDEB has C_{2h} symmetry including an inversion center. The increased symmetry of PDEB makes the C=C stretching bands forbidden in the 1610–1570 cm^{-1} region. Actually, the calculated intensities of the 1605 and 1573 cm^{-1} bands are both zero for PDEB.

On the basis of those C=C stretching band patterns, the spectral differences in the 2100–2220 nm region can be clearly understood. In the case of PDEB, the strong band is observed at 2172 nm (4604 cm^{-1}), which results from the second strong C=C stretching band at 1516 cm^{-1}. On the other hand, the combination bands of ODEB and MDEB near 2174 nm (4600 cm^{-1}) are weak due to the relatively weak intensity of the C=C stretching band at 1490 cm^{-1} as well as the influence of bands in both the 1465–1450 and 1610–1570 cm^{-1} regions. From both experimental results and theoretical considerations, it is believed that the C=C stretching band patterns change with different symmetry types of *ortho*-, *meta*-, and *para*-substituted positions, and they also produce the spectral differences of combination NIR bands in the 2100–2220 nm region.

Conversely, the spectral features of PX differ from those of diethylbenzene isomers, especially in the 2300–2400 and 2430–2500 nm ranges (see Fig. 1). The different molecular structure of PX, which contains only =C–H and C=C vibrations, produces noticeable spectral features. Overall, the absorption bands of four components (ODEB, MDEB, PDEB, and PX) have unique and selective spectral features, especially in the combination NIR region.

Figure 3 shows the 40 randomly selected NIR spectra of samples in the calibration set. The 2100–2500 nm range is magnified in the same plot. As shown, the spectral variations are large because the concentrations of MDEB, PDEB, and PX are changing widely, as shown in Table I. As separation proceeds, the concentration of PDEB increases while the concentration of PX decreases. NIR spectral features clearly follow this trend. The die-

TABLE II. The compositions of three selected samples (designated as A, B, and C in Fig. 4) (volume %).

	ODEB	MDEB	PDEB	PX
Sample A	2.33	12.51	76.15	1.09
Sample B	3.70	56.27	23.20	13.19
Sample C	0.11	0.91	10.48	87.88

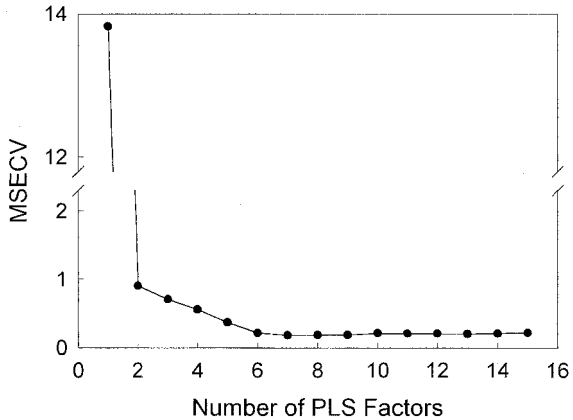

FIG. 5. The effect of number of PLS factors on MSECV plotted as a function of the number of PLS factors used to determine the PDEB content within the 2100–2500 nm region.

thylbenzene band at 2296 nm is increasing while the PX band at 2320 nm is decreasing as the separation proceeds. The selective spectral information among diethylbenzene isomers is located in the 2100–2220 and 2430–2500 nm ranges, as discussed.

PCA Analysis. The absorption bands of all the components are relatively poorly resolved compared to those in the MIR and have similar spectral features. Unique spectral features of each component, however, can provide selective information for a quantification using multivariate calibration techniques such as a factor-based analysis. NIR spectra of mixture samples are highly overlapped; therefore, multivariate calibration methods are required to resolve these mixtures, as has been determined empirically in the field. To initially examine compositional changes more clearly, PCA (principal component analysis) was performed on the calibration data set along with pure spectra of each component. The main purpose was to compare the scores of calibration and pure samples simultaneously. Figure 4 shows the score scatter plot using first and second principal components. The scores of each pure component are present as diamond (ODEB), circle (MDEB), square (PDEB), and triangle (PX). The scores of MDEB, PDEB, and PX form a triangle. The scores of calibration samples are located inside of this triangle, because the samples are mainly composed of MDEB, PDEB, and PX.

To investigate how the scores of samples were clearly following concentration variation, we selected three samples (designated as A, B, and C in Fig. 4) near the scores of each component. The compositions of these samples are presented in Table II. For sample A, which contains the highest concentration of PDEB in the calibration data set, the score is closest to that of PDEB. Similarly, sample B contains the highest concentration of MDEB, and the corresponding score is also closest to that of MDEB. Sample C contains a fairly high concentration of PX, and its score is close to that of PX. Consequently, the concentration variations are clearly identified by using principal components (factors).

PLS Calibration Models. Partial least-squares regression was used to build calibration models for each component. In the application of the PLS algorithm, it is generally known that the spectral range and number of PLS factors are critical parameters. The spectral range determines the location and quality of spectral information, and the number of PLS factors should be optimally selected to avoid overfitting. It has been previously determined that the calibration result depends on the spectral region used.[13] Four different spectral regions of 1100–2500, 1600–2500, 2100–2500, and 1100–1800 nm were tested. These regions were selected to include combinations of the four major absorption bands (1100–1300, 1300–1450, 1650–1800, and 2100–2500 nm). The 1100–2500 range incorporates all four absorption bands, and the 1600–2500 nm range uses the combination (2100–2500 nm) and first-overtone (1650–1800 nm) bands. The 2100–2500 nm uses only the combination bands that are the most sensitive, while the 1100–1800 nm range uses the less sensitive first-overtone, combination (1300–1450), and second-overtone bands.

The optimum number of factors was identified as the number of factors that give a minimum MSECV (mean square error of cross-validation). The cross-validation involved dividing the calibration data set into seven segments. The MSECV is a good index to find the optimal number of PLS factors without overmodelling. Figure 5 shows the MSECV plotted as a function of the number of PLS factors used to determine the PDEB content within the 2100–2500 nm band. As expected, the MSECV decreases very sharply for the first factor and gradually decreases as more PDEB-dependent spectral variation is incorporated into the calibration model. As the number of factors increase further, the MSECV begins to increase at eight PLS factors. An increase in the MSECV indicates that the data have been over-fit by incorporating spectral information into the model that is not related to PDEB. In this case, seven PLS factors were chosen as optimum. The optimum numbers of PLS factors for other spectral regions and other components were determined by generating and examining the same types of plots.

The calibration results for each component with the use of raw (i.e., log $1/R$) spectra are summarized in Table III. The standard error of prediction (SEP) was calculated by predicting the samples in the confirmation data set. As shown in Table III, when the combination band of 2100–

TABLE III. PLS calibration results for each component using log $1/R$ spectra (unit: vol %).[a]

Range (nm)	ODEB		MDEB		PDEB		PX	
	MSECV	SEP	MSECV	SEP	MSECV	SEP	MSECV	SEP
1100–2500	0.11 (6)	0.11	0.15 (8)	0.16	0.16 (8)	0.15	0.27 (8)	0.27
1600–2500	0.11 (6)	0.11	0.15 (8)	0.16	0.17 (7)	0.17	0.27 (9)	0.28
2100–2500	0.11 (6)	0.12	0.15 (8)	0.16	0.16 (7)	0.15	0.28 (7)	0.27
1100–1800	0.13 (5)	0.15	0.25 (8)	0.22	0.27 (6)	0.25	0.28 (7)	0.27

[a] Numbers in parentheses correspond to the number of PLS factors used.

TABLE IV. PLS calibration results for each component using second-derivative spectra (unit: vol %).[a]

Range (nm)	ODEB		MDEB		PDEB		PX	
	MSECV	SEP	MSECV	SEP	MSECV	SEP	MSECV	SEP
1100–2500	0.10 (5)	0.11	0.14 (8)	0.14	0.17 (6)	0.15	0.26 (8)	0.29
1600–2500	0.10 (5)	0.11	0.14 (7)	0.15	0.17 (6)	0.15	0.26 (8)	0.29
2100–2500	0.11 (5)	0.11	0.14 (7)	0.15	0.16 (7)	0.14	0.26 (8)	0.29
1100–1800	0.12 (6)	0.16	0.24 (8)	0.20	0.27 (6)	0.22	0.28 (5)	0.28

[a] Numbers in parentheses correspond to the number of PLS factors used.

2500 nm (the most sensitive and selective region) is included, the calibration and prediction results for three isomers are similar whether other spectral ranges are included or not. On the contrary, when only the 1100–1800 nm range (without the 2100–2500 nm range) is used, the calibration results are considerably degraded. These results demonstrate that the most valuable and selective information among diethylbenzene isomers is located in the 2100–2500 nm region, as expected. For PX, the calibration results are similar regardless of spectral range, since the spectral features of PX are clearly different from diethylbenzene isomers over the NIR region.

The same PLS algorithm was used on second-derivative spectra. Figure 3 (bottom plot) shows the second-derivative spectra of 40 randomly selected samples in the calibration set. The spectral variations are effectively enhanced while keeping the same qualitative information. The results of calibration employing second-derivative spectra are summarized in Table IV. As shown in the table, calibration results are similar to those of the log $1/R$ spectra. The baseline variation is not significant, and spectral variation is clearly observable in the log $1/R$ spectra; therefore, no significant improvement in calibration is achieved by using second-derivative spectra.

Scatter plots showing correlation between NIR and GC analyses (with the use of the 2100–2500 nm range of log $1/R$ spectra) for each component are shown in Fig. 6. Open and filled circles represent calibration and confirmation data, respectively. In all cases, the calibration and confirmation data show excellent correlation with conventional GC data, and many points fall on or close to the unity line. Concentrations in the confirmation data set were accurately predicted with SEPs of 0.11 vol % for ODEB, 0.16 vol % for MDEB, 0.15 vol % for PDEB, and 0.27 vol % for PX.

CONCLUSION

NIR spectroscopy has been successfully implemented and has replaced the conventional GC for compositional analysis of PDEB in the separation process. Prediction results of each NIR calibration model developed with PLS regression showed an excellent correlation with the GC data. The characteristic C=C ring stretches of each diethylbenzene isomers in the MIR region (1600–1450 cm^{-1}) produced selective and identifiable spectral features in the NIR region, especially in the combination region at 2100–2220 nm. With NIR spectroscopy, the major components in the PDEB separation process can be measured in less than one minute. With the help of fast NIR analysis, the process status can be monitored in real time, and necessary on-line process control and optimization

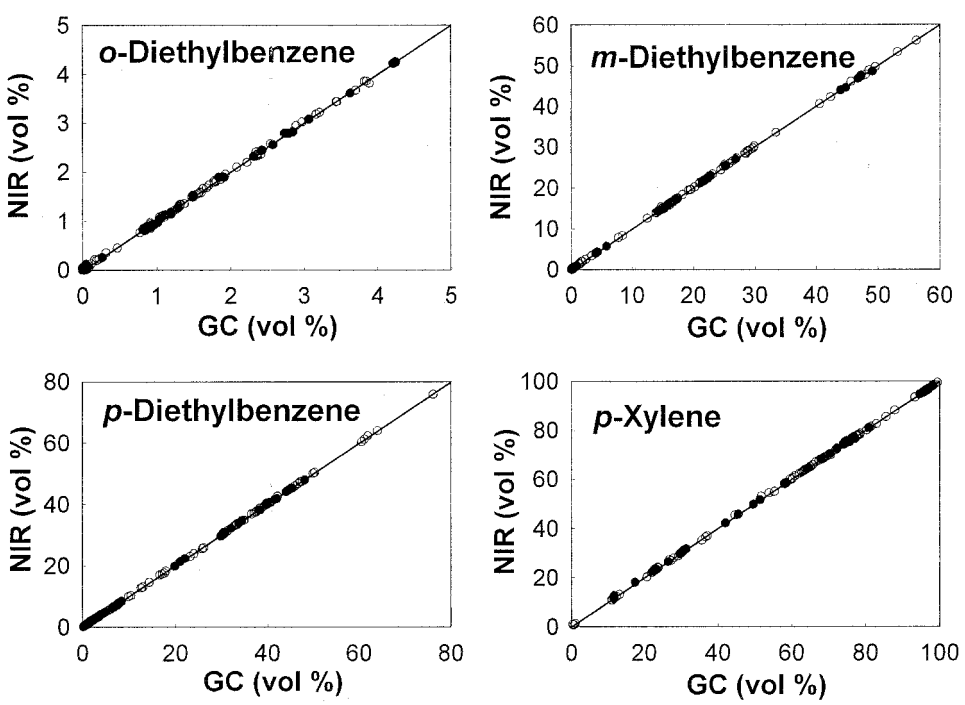

FIG. 6. Scatter plots showing correlation between NIR and GC analyses (using the 2100–2500 nm range of log $1/R$ spectra) for each component.

methods can be easily combined. Future work will evaluate the on-line process NIR analyzer system including a sample conditioning unit for real-time process control and optimization.

ACKNOWLEDGMENT

Dr. Jaebum Choo thanks the Korea Research Foundation for financial support (Grant No. 1999-015-DI0053).

1. T. Lu and T. Lee, Ind Eng Chem Rev **26,** 2024 (1987)
2. M. Mazzotti, R. Baciocchi, G. Storti, and M. Morbidelli, Ind. Eng. Chem. Rev. **35,** 2313 (1996).
3. D. M. Ruthven, *Principles of Adsorption & Adsorption Processes* (Wiley–Interscience, New York, 1984), p. 396.
4. M. Suzuki, *Adsorption Engineering* (Elsevier, Amsterdam, 1990), p. 238.
5. D. A. Burns and E. W. Ciurczak, *Handbook of Near-Infrared Analysis* (Marcel Dekker, New York, 1992).
6. D. L. Wetzel, Anal. Chem. **55,** 1165A (1983).
7. C. Lee, W. Yang, and R. G. Parr, Phys. Rev. **B37,** 785 (1988).
8. A. D. Becke, J. Chem. Phys. **98,** 5648 (1993).
9. M. J. Frisch, A. Frisch, and J. M. Foresman, *Gaussian 94 User's Reference* (Gaussian Inc., Pittsburgh, Pennsylvania, 1995).
10. H. Martens and T. M. Næs, *Multivariate Calibration* (John Wiley and Sons, New York, 1989), p. 116.
11. K. R. Beebe, R. J. Pell, and M. B. Seasholtz, *Chemometrics: A Practical Guide* (John Wiley and Sons, New York, 1998).
12. M. J. Frisch, G. W. Trucks, H. B. Schlegel, P. M. W. Gill, B. G. Johnson, M. A. Robb, J. R. Cheeseman, T. Keith, G. A. Petersson, J. A. Montgomery, K. Raghavachari, M. A. Al-Laham, V. G. Zakrzewski, J. V. Ortiz, J. B. Foresman, J. Cioslowsski, B. B. Stefanov, A. Nanayakkara, M. Challacombe, C. Y. Peng, P. Y. Ayala, W. Chen, M. W. Wong, J. L. Andres, E. S. Replogle, R. Gomperts, R. L. Martin, D. J. Fox, J. S. Binkley, D. J. Defrees, J. Baker, J. P. Stewart, M. Head-Gordon, C. Gonzalez and J. A. Pople, *Gaussian 94, Revision D.2* (Gaussian Inc., Pittsburgh, Pennsylvania, 1995).
13. H. Chung, J. S. Lee, and M. S. Ku, Appl. Spectrosc. **52,** 885 (1998).